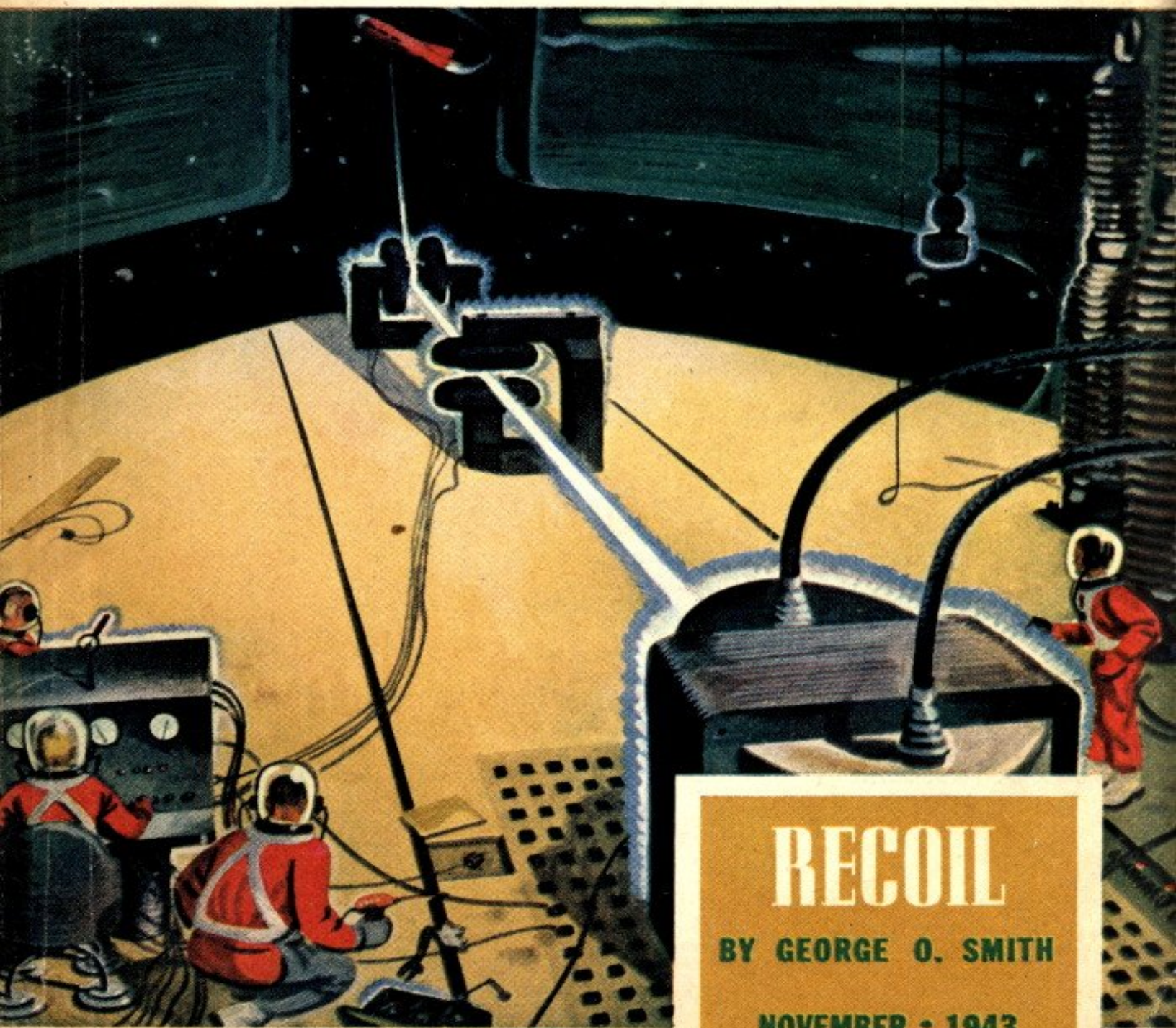


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Science Fiction



RECOIL

BY GEORGE O. SMITH

NOVEMBER • 1943

25 CENTS

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*** START OF THE PROJECT GUTENBERG EBOOK RECOIL ***

Recoil

By George O. Smith

Illustrated by Orban

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Walter Franks sat in the director's office; his feet on the director's desk. He was smoking one of the director's cigarettes. He was drinking the director's liquor, filched shamelessly from the director's private filing cabinet where it reposed in the drawer marked "S." Drawer "B" would have given beer, but Walt preferred Scotch.

He leaned forward and tossed the director's cigarette into the director's wastebasket and then he pressed the button on the desk and looked up.

But it was not the director's secretary who entered. It was his own, but that did not disturb Franks. He knew that the director's secretary was off on Mars enjoying a honeymoon with the director.

Jeanne entered and smiled. "Must you call me in here to witness you wasting the company's time?" she asked in mock anger.

"Now look, Jeanne, this is what Channing does."

"No dice. You can't behave as Don Channing behaves. The reason is my husband."

"I didn't call to have you sit on my lap. I want to know if the mail is in."

"I thought so," she said. "And so I brought it in with me. Anything more?"

"Not until you get a divorce," laughed Franks.

"You should live so long," she said with a smile. She stuck her tongue out at him.

Walt thumbed his way through the mail, making notations on some, and setting others aside for closer reading. He came to one and tossed it across the desk at Jeanne. She took the message and read:

Dear Acting Director:

Having a wonderful honeymoon; glad you aren't here!

Don and Arden.

"Wonderful stuff, love," smiled Franks.

"It is," agreed Jeanne. A dreamy look came into her eyes.

"Scram, Jeanne. There are times when you can't work worth a darn. Usually when you're thinking of that husband of yours. What's he got that I haven't?"

"Me," said Jeanne slyly. She arose and started for the door. "Oh," she said, "I almost forgot. Warren phoned up and said that the turret is ready for a try-out."

"Fine," said Walt. "Swell." He unfolded himself from the chair with alacrity and almost beat the girl to the door.

"My," she laughed, "you can move after all."

"Sure," he grinned. "Now I have something for which to live."

"I hope it's worth it. You've sunk a lot of change into that bug-house."

"I know, but we can stand it. After all, since Don took over this affair, Interplanetary Communications is an up and running business. We're out of the Government subsidy class now, and are making money. If this works, we'll make more. It's worth a gamble."

"What are you trying to build?" asked Jeanne.

"Why, since this business of contacting ships-at-space has become so universally liked, we have a tough time keeping ships in the mobile beam. That's because they are always ducking out of the way of loose meteorites and stuff, and that screws up their course. We can't see 'em, and must take their position on the basis of their expected course. We never know whether we hit 'em until they land.

"Now I've been trying to devise a space gun that will blast meteors directly instead of avoiding them by coupling the meteor detector to the autopilot."

"Gonna shoot 'em out of existence?"

"Not exactly. Popping at them with any kind of a rifle would be like trying to hit a flying bird with a spitball. Look, Jeanne, top speed on the run from Mars to Terra at major opposition is up among the thousands of miles per second at the turnover. A meteor itself may be blating along at fifty miles per second. Now a rifle, shooting a projectile at a few thousand feet per second would be useless. You'd have the meteor in your lap and out of the other side while the projectile is making up its mind to move forward and relieve the pressure that is building up behind it due to the exploding powder.

"I've designed an electron gun. It is a superpowered, oversized edition of the kind they used to use in kinescope tubes, oscilloscope tubes, and electron microscopes. Since the dingbat is to be used in space, we can leave the works of the gun open and project a healthy stream of electrons at the offending object without their being slowed and dispersed by an impending atmosphere."

"But that sounds like shooting battleships with a toy gun."

"Not so fast on the objections, gal," said Franks. "I've seen a simple oscilloscope tube with a hole in the business end. It was burned right through a quarter inch of glass because the fellows were taking pix and had the intensity turned up high. The sweep circuit blew a fuse and the beam stopped on one spot. That was enough to puncture the screen."

"I see. That was just a small affair?"

"A nine-inch tube. The electron gun in a nine-inch kinescope tube is only about four inches long and three quarters of an inch in diameter. Mine, out there in the turret, is six feet in diameter and thirty feet long. I can fire out quite a bundle of electrons from a tube of that size."

"It sounds as though you mean business."

"I do. This is the right place to do research of that kind. Out here on Venus Equilateral, we're in a natural medium for an electron gun, and we've the power requirements to run it. I can't think of any place in the System that offers better chances."

"When are you going to try it out?"

"As soon as a meteor comes over the pike, as long as Warren says we're ready."

Jeanne shook her head. "I wish Channing were here. Things are wild enough when you are both working on something screwball, but I could get scared something fierce at the thought of either one of you working without the other."

"Why?"

"You two sort of act as balance wheels to one another's craziness. Oh, don't take that word to heart. Everybody on the Relay Station thinks the world of you two, myself included. *Craziness* in this case means a sort of friendly

description of the way your brains work. Both of you dash off on tangents now and then, and when either one of you get off the beam, the other one seems to swing the weight required to bring the lost one back to the fold."

"That's a real mess of mixed metaphors, Jeanne. But I am going to surprise Don hairless when he gets back here and finds that I've done what people claimed couldn't be done. I'm going to be the bird whose bust sits in the Hall of Fame in between Edison, Einstein, Alexander Graham Bell, S. F. B. Morse, and—"

"Old Man River, Jack Frost, and Little Boy Blue," laughed Jeanne. "I hope it's not a bust, Walt."

"You mean I should have a whole statue?"

"I mean, I hope your dream is not a bust!"

Jeanne left, with Walt right behind her. Franks did not remain at the desk, however, but made his way from the office level to the outer skin of the Relay Station by way of a not-often-used stairway that permitted him to drop to the outer skin. Above his head were the first levels of apartmental cubicles occupied by the personnel of Venus Equilateral. Out here, Walt had but a scant thickness of steel between him and the void of space.

His pathway was strewn with pipe, cable, and storage tanks. He passed a long-forgotten project and paused to reminisce over the days when a meteor shower had caused them some concern by puncturing the skin twice. The installation of a sponge elastomer under compression in this space had been stopped when a brilliant astrophysicist proved to Channing—then a supervisor in the operations laboratory—that the chances of being dangerously punctured were practically nil, and that the actual puncturing had done nothing but make people uncomfortably leery.

Then Franks came to a room built from outer skin to inner skin and about fifty feet in diameter. He unlocked the door with a key on his watch chain, and entered. Jim Warren was waiting for him.

"Hi, ordnance expert. We're ready as soon as they are."

"How's she working?"

"I should know? We've been squirting ropes of electrons out to blank space for hours. She gets rid of them all right. But have we done any good? I dunno."

"Not a meteor in sight, I suppose."

"The detector hasn't blinked once. But when she does, your electron gun will follow the darned thing until it gets a half thousand miles out of sight, or will pick it up a thousand miles before it gets here."

"That sounds fine. It's a good thing that we don't have to swivel that mess of tube around a whole arc in actual use. It would take too long. But we'll put one in each upper quadrant of a spaceship and devise it so that its working arc will be small enough to make it work. Time enough to find that out after we know if it works."

"That's something that I've been wondering about," said Warren. "Why didn't we build a small one out here and evacuate the skin for a few hundred feet? We could set up a few chunks of iron and squirt electrons at 'em."

"And have the folks upstairs screaming? Nope. I've a hunch that when this beam hits something hard, it will create quite a ruckus. It would be fine to have a hunk blown right out of the skin, wouldn't it?"

"Guess you're right," admitted Warren.

The meteor alarm flashed, and a bell dinged once.

"Here's our chance," snapped Walt. "We've about fifteen seconds to work on this one."

He looked out of a tiny window, and saw that the big tube had lined up with the tiny model that was a monitor for the big tube. He sighted through the model, which in itself was a high-powered telescope, and he saw the jagged meteor rushing forward at an angle to the Station. It would miss by miles, but it would offer a good target.

"Cathode's hot," said Warren.

Walt Franks grasped the power switch and thrust it down part way. Meters leaped up their scales and from somewhere there came the protesting whine of tortured generators. Through the window, nothing very spectacular was

happening. The cathode glowed slightly brighter due to the passage of current through its metal and out of the coated surface. But the electrostatic stresses that filled the gaps between the accelerator and focusing anodes was no more visible than the electricity that runs a toy motor. Its appearance had not changed a bit, but from the meters, Walt Franks knew that megawatts of electronic power, in the shape of high-velocity electrons, was being poured from the cathode; accelerated by the ring anodes; and focused to a narrow beam by the focusing anodes. And from the end of the framework that supported these anodes, a cylinder of high-velocity electrons poured forth, twelve inches in diameter.

Through the telescope, the meteor did not seem to be disturbed. It exploded not, neither did it melt. It came on inexorably, and if the inanimate nickel and iron of a meteor can be said to have such, it came on saucily and in utter disregard for the consequences.

Frantically, Walt cranked the power up higher and higher, and the lights all over the Station dimmed as the cathode gun drained the resources of the Station.

Still no effect.

Then in desperation, Walt slammed the power lever down to the bottom notch. The girders strained in the tube from the terrific electrostatic stresses, and for a second, Walt was not certain that the meteor was not finally feeling the effects of the electron bombardment.

He was not to be sure, for the experiment came to a sudden stop.

An insulator arced where it led the high-voltage lines that fed the anodes through the wall. Immediately it flashed over, and the room filled to the brim with the pungent odor of burning insulation. A medium-voltage anode shorted to one of the high-voltage anodes, and the stress increased in the tube. It broke from its moorings, this low-voltage anode, and it plunged backward, down the tube toward the cathode. It hit, and it was enough to jar the whole tube backward on its gimbals.

The shock warped the mounting of the tube, and it flexed slightly, but sufficiently to bring the farthestmost and highest voltage anode into the electron stream. It glowed redly, and the secondary emission raved back through the series of electrodes, heating them and creating more warpage.

Then the pyrotechnic stopped. Great circuit breakers crashed open up in the power room hundreds of feet above them, high in the Station.

Walt Franks looked out through the window at the tangled mess that had been a finely machined piece of equipment. He saw the men looking quizzically at him as he turned away from the window, and with a smile that cost him an effort, he said: "All right, so Marconi didn't get WLW on his first try, either. Come on, fellows, and we'll clean up this mess."

With the utter disregard that inanimate objects show toward the inner feelings of the human being, the meteor alarm blinked again and the bell rang. The pilot tube swiveled quickly to one side, lining up with the spot in the celestial globe of the meteor detector. Out in the turret that housed the big tube, motors strived against welded commutators and the big tube tried to follow.

Walt looked at the pointing tube and said: "All right! Go ahead and point!"

Don Channing smiled at Arden. "Mrs. Channing," he said, "must you persist in keeping me from my first love?"

Arden smiled winningly. "Naturally. That's what I'm here for. I intend to replace your first love entirely and completely."

"Yeah," drawled Don, "and what would we live on?"

"I'll permit you to attend to your so-called first love during eight hours every day, providing that you remember to think of me every half-hour."

"That's fine. But you really aren't fair about it. We were on Terra for two weeks. I was just getting interested in a program outlined by one of the boys that works for Interplanet, and what happened? You hauled me off to Mars. We stayed for a week at the Terraland Hotel at Canalopsis and the first time that Keg Johnson came to see us with an idea and a sheaf of papers, you rushed me off to Lincoln Head. Now I'm scared to death that some guy will try to open a blueprint here; at which I'll be rushed off to the Palanortis Country until someone finds us there. Then it'll be the Solar Observatory on Mercury or the Big Glass on Luna."

Arden soothed Don's feelings by sitting on his lap and snuggling. "Dear," she said in a voice that positively dripped, "we're on a honeymoon, remember?"

Don stood up, dumping Arden to the floor. "Yeah," he said, "but this is the highest velocity honeymoon that I ever took!"

"And it's the first one I was ever on where the bridegroom took more time admiring beam installations than he took to whisper sweet nothings to his gal. What has a beam transmitter got that I haven't got?"

"One: Its actions can be predicted. Two: It can be controlled. Three: It never says anything original, but only repeats what it has been told. Four: It can be turned off."

Arden caught Don on the point of the chin with a pillow and effectively smothered him. She followed her slight advantage with a frontal attack that carried him backward across the bed, where she landed on top viciously and proceeded to lambaste him with the other pillow.

It was proceeding according to plan, this private, good-natured war, until a knock on the door caused a break in operations. Channing struggled out from beneath Arden and went to the door trying to comb his hair by running spread fingers through it. He went with a sense of failure caused by Arden's quiet laugh and the statement that he resembled a bantam rooster.

The man at the door apologized, and then said: "I'm Doug Thomas of the *Triworld News*."

"Come in," said Don, "and see if you can find a place to sit."

"Thanks."

"I didn't know that *Triworld News* was interested in the wedded life of the Channings. Why doesn't *Triworld* wait until we find out about it ourselves?"

"*Triworld* does not care to pry into the private life of the newly wed Channing family," laughed Doug. "We, and the rest of the System do not give a damn whether Mrs. Channing calls you Bunny-bit or Sugar-pie—"

"Sweetums," corrected Arden with a gleam in her eye.

"—we've got something big to handle. I can't get a thing out of the gang at Canalopsis, they're all too busy worrying."

"And so you came here? What do you expect to get out of us? We're not connected in any way with Canalopsis."

"I know," said Doug, "but you do know space. Look, Channing, the *Solar Queen* has been missing since yesterday morning!"

Don whistled.

"See what I mean? What I want to know is this: What is your opinion on the matter? You've lived in space for years, on the Relay Station, and you've had experience beyond anybody I can reach."

"Missing since yesterday morning," mused Channing. "That means trouble."

"That's what I thought. Now if you were running the spaceport at Canalopsis, what would your own private opinion be?"

"I don't know whether I should speak for publication," said Don.

"It won't be official. I'll corroborate anything you say before it is printed, and so on. But I want an unofficial opinion, too. If you want this withheld, say so, but I still want a technical deduction to base my investigation on. I don't understand the ramifications and the implications of a missing ship. It is enough to make Keg Johnson's hair turn gray overnight, though, and I'd like to know what is so bad before I start to turn stones."

"Well, keep it off the record until Canalopsis gives you the go-ahead. I can give you an opinion, but I don't want to sound official."

"O.K. Do you suppose she was hit by a meteor shower?"

"Doubt it like the devil. Meteor detectors are many and interconnected on a spaceship, as well as being alarmed and fused to the nth degree. Any trouble with them will bring a horde of ringing bells all through the ship which would bring the personnel a-running. They just don't go wrong for no reason at all."

"Suppose that so many meteors came from all directions that the factors presented to the autopilot—"

"No dice. The possibility of a concentration of meteors from all directions all about to pass through a certain spot in space is like betting on two Sundays in a row. Meteors don't just run in all directions, they have a general drift. And the meteor detecting equipment would have been able to pick up the centroid of any group of meteors soon enough to lift the ship around it. Why, there hasn't been a ship hit by a meteor in ten years."

"But—"

"And if it had been," continued Channing, "the chances are more than likely that the ship wouldn't have been hit badly enough to make it impossible to steer, or for the crew to shoot out message tubes which would have landed on Canalopsis."

"Suppose that the ship ducked a big shower and it went so far out of course that they missed Mars?"

"That's out, too," laughed Channing.

"Why?"

"A standard ship of space is capable of hitting it up at about 4-G all the way from Terra to Mars at major opposition and end up with enough power and spare cathodes to continue on to Venus in quadrature. Now the velocity of the planets in their orbits is a stinking matter of miles per second, while the top speed of a ship in even the shortest passage runs up into four figures per second. You'd be surprised at what velocity you can attain at 1-G for ten hours."

"Yes?"

"It runs to slightly less than two hundred and fifty miles per second, during which you've covered only four million miles. In the shortest average run from Venus to Terra at conjunction, a skimpy twenty-five million miles, your time of travel is a matter of twenty-five hours add, running at the standard 2-G. Your velocity at turnover—or the halfway point where the ship stops going up from Terra and starts to go down to Venus—is a cool five hundred miles per second. So under no condition would the ship miss its objective badly enough to cause its complete loss. Why, this business is run so quickly that were it not for the saving in time and money that amounts to a small percentage at the end of each flight, the pilot could head for his planet and approach the planet asymptotically."

"You know what you're doing, don't you?" asked the reporter.

"I think so."

"You're forcing my mind into accepting something that has never happened before, and something that has no basis for its—"

"You mean piracy? I wonder. We've all read tales about the Jolly Roger being painted on the side of a sleek ship of space while the pirate, who at heart is a fine fellow though uninhibited, hails down the cruiser carrying radium. He swipes the stuff and kisses all the women whilst menacing the men with a gun hand full of searing, coruscating, violently lethal ray pistol. But that sounds fine in stories. The trick is tougher than it sounds, Thomas. You've got to catch your rabbit first."

"Meaning?"

"Meaning that finding a ship in space to prey upon is somewhat less difficult than juggling ten billiard balls whilst riding a horse blind-folded. Suppose you were to turn pirate. This is what would happen:

"You'd get the course of the treasure ship from the spaceport, fine and good, by resorting to spies and such. You'd lie in wait out there in the blackness of space, fixing your position by the stars and hoping that your error in fix was less than a couple of thousand miles. It's more likely to be a hundred thousand miles, though. The time comes. You look to your musket, sharpen your sword, and see to the priming of your Derringers that are thrust into the red sash at your waist. You are right on the course, due to your brilliant though lawless navigator who was tossed out of astrogator's school for filching the teacher's whiskey. Then the treasure ship zoops past at a healthy hundred miles per second and you decide that since she is hitting it up at 2-G, you'd have to start from scratch at a heck of a lot better to catch her within the next couple of light years.

"So you give up, join the Congregational Church and pass the collection plate every Sunday."

"But suppose you took the course as laid and applied the same acceleration? Suppose you followed on the heels of your quarry until you were both in space? You could do it then, couldn't you?"

"Gosh," said Channing, "I never thought of that. That's the only way a guy could pirate a ship—unless he planted his men aboard and they mutinied."

"Then it might be pirates?"

"It might be," admitted Channing. "It'd have to occur near beginning or end, of course, though. I can't think of anything safer than being shot at out of a gun of any kind while both crates are hitting it up at a couple of hundred miles per second and at a distance of a few miles apart. It would be all right if you were both running free, but at 2-G acceleration, you'd have to do quite a bit of ballistic gymnastics to score a hit."

"Or run in front of your quarry and sow a bouquet of mines."

"Except that the meteor detector would show the position of the pirate craft in the celestial globe and the interconnecting circuits would cause the treasure ship to veer off at a sharp angle. Shucks, Doug, this thing has got too many angles to it. I can't begin to run it off either way. No matter how difficult it may sound, there are still ways and means to do it. The one thing that stands out like a sore thumb is the fact that the *Solar Queen* has turned up missing. Since no inanimate agency could cause failure, piracy is the answer."

"You're sure of that?"

"Not positive. There are things that might cause the ship to founder. But what they are depend upon too many coincidences. It's like hitting a royal flush on the deal, or filling a full house from two pair."

"Well, thanks, Channing. I'm heading back to Canalopsis right now. Want to come along?"

Channing looked at Arden, who was coming from the dressing room carrying her coat and he nodded. "The gal says yes," he grinned. "Annoy her until I find my shoes, will you?"

Arden wrinkled her nose at Don. "I'll like that," she said to Doug.

The trip from Lincoln Head to Canalopsis was a fast one. Doug Thomas drove the little flier through the thin air of Mars at a breakneck speed and covered the twelve hundred miles in just shy of two hours. At the spaceport, Channing found that he was not denied the entrance as the reporter had been. He was ushered into the office of Keg Johnson in record time, and the manager of the Canalopsis Spaceport greeted Don with a worried expression on his face.

"Still gone," he said cryptically. "Like the job of locating her?"

Don shook his head with a sympathetic smile. "Like trying to find a grain of sand on a beach—a specified grain, I mean. Wouldn't know how to go about it."

Keg nodded. "I thought as much. That leaves her out of the picture. Well, up to now space travel has been about as safe as spending the evening in your easy-chair. Hello, Arden, how's married life?"

"Can't tell yet," she said with a twinkle. "I've got to find out whether I can break him of a dozen bad habits before I'll commit myself."

"I wish you luck, Arden, although from that statement, it's Don that needs the luck."

"We came to see if there was anything we could do about the *Solar Queen*," offered Channing.

"What can anybody do?" asked Keg with spread hands. "About all we can do is to put it down in our remembrances and turn to tomorrow. Life goes on, you know," said Keg in a resigned tone, "and either we keep up or we begin to live in the past. Are you going to stay here for a day or two?"

"Was thinking about it," said Don.

"Well, suppose you register at the Terraland and meet me back here for lunch. If anything occurs, I'll shoot you a quickie." Keg looked at his watch and whistled. "Lordy," he said ruefully. "I didn't know how late it was. Look, kids, I'll run you downtown myself, and we'll all have lunch at the Terraland. How's that?"

"Sounds better," admitted Channing. "My appetite, you know."

"I know," laughed Arden. "Come on, meat-eater, and we'll peel a calf."

It was during lunch that a messenger raced into the dining room and handed Keg a letter. Keg read, and then swore roundly. He tossed the letter across the table to Don and Arden.

To the Operators of all Spacelines:

It has come to my attention that your ships require protection. The absence of the *Solar Queen* is proof enough that your efforts are insufficient to insure the arrival of a spaceship at its destination.

I am capable of offering protection at the reasonable rate of one dollar solarian for every gross ton, with the return of ten dollars solarian if any ship fails to come through safely. I think that you may find it necessary to subscribe to my insurance, since without my protection I cannot be responsible for failures.

ALLISON (HELLION) MURDOCH.

"Why the dirty racketeer," stormed Arden. "Who is he, anyway?"

"Hellion Murdoch is a man of considerable ability as a surgeon and a theoretical physicist," explained Don. "He was sentenced to the gas chamber ten years ago for trying some of his theories out on human beings without their consent. He escaped with the aid of fifteen or twenty of his cohorts who had stolen the *Hippocrates* right out of the private spaceport of the Solarian Medical Research Institute."

"And they headed for the unknown," offered Keg. "Wonder where they've been for the last ten years."

"I'll bet a hat that they've been in the Melapalan Jungle, using the machine shop of the *Hippocrates* to fashion guns. That machine shop was a dilly, if I remember correctly."

"It was. The whole ship was just made to be as self-sustaining as it could be. They used to run all over the System in it, you know, chasing bugs. But look, Don, if I were you, I'd begin worrying about Venus Equilateral. That's where he'll hit next."

"You're right. But what are you going to do?"

"Something that will drive him right out to the Relay Station," said Keg in a sorrowful tone. "Sorry, Don, but when I put an end to all space shipping for a period of six weeks, Hellion Murdoch will be sitting in your lap."

"He sure will," said Channing nervously. "Arden, are you willing to run a gantlet?"

"Sure," she answered quickly. "Are you sure that there will be danger?"

"Not too sure, or I wouldn't take you with me. Unless Murdoch has managed to build himself a couple of extra ships, we've got a chance in three that he'll be near one of the other two big spaceports. So we'll slide out of here

unannounced and at a peculiar time of day. We'll load up with gravanol and take it all the way to the Station at 6-G."

"He may have two or three ships," said Keg. "A man could cover all the standard space shipping in three, and he might not have too bad a time with two, especially if he were only out looking for those which weren't paid for. But, look, I wouldn't check out of the Terraland if I were you. Keep this under cover. Your heap is all ready to take sky from Canalopsis Spaceport and you can leave directly."

"Hold off on your announcement as long as possible," Don asked Keg.

Johnson smiled and nodded. "I'll give you time to get there anyway. But I've no control over what will be done at Northern Landing or Mojave. They may kick over the traces."

"Arden, we're moving again," laughed Don. "Keg, ship us our duds as soon as this affair is cleared up." Channing scribbled a message on the back of Murdoch's letter. "Shoot this off to Walt Franks, will you? I won't wait for an answer, that'll take about fifty minutes, and by that time I'll have been in space for twenty."

They paused long enough to stop at the nurse's office at the spaceport for a heavy shot of gravanol and a thorough bracing with wide adhesive tape. Then they made their way to the storage space of the spaceport where they entered their small ship. Channing was about to send the power lever home when the figure of Keg Johnson waved him to stop.

Keg ran up to the space lock and handed in a paper.

"You're it," he said. "Good luck, Channings."

It was another message from Hellion Murdoch. It said, bluntly:

To Donald Channing,
Director of Communications:

Considerable difficulty has been experienced in transmitting messages to the interested parties. I desire a free hand in telling all who care, the particulars of my insurance.

Since your Relay Station is in a position to control all communications between the worlds, I am offering you the option of either surrendering the Station to me, or of fighting me for its possession. I am confident that you will see the intelligent course; an unarmed station in space is no match for a fully armed and excellently manned cruiser.

Your answer will be expected in five days.

ALLISON (HELLION) MURDOCH.

Channing snarled and thrust the power lever down to the last notch. The little ship leaped upward under 5-1/2-G, and was gone from sight in less than a minute.

Arden shook her head. "What was that message you sent to Franks?" she asked.

"I told him that there was a wild-eyed pirate on the loose, and that he might make a stab at the Station. We are coming in as soon as we can get there and to be on the lookout for us on the landing communications radio, and also for anything untoward in the nature of space vessels."

"Then this is not exactly a shock," said Arden, waving the message from Murdoch.

"Not exactly," said Channing dryly. "Now look, Arden, you go to sleep. This'll take hours and hours, and gabbing about it will only lay you out cold."

"I feel fine," objected Arden.

"I know, but that's the gravanol, not you. The tape will keep you intact, and the gravanol will keep you awake without pain or nausea. But you can't get something for nothing, Arden, and when that gravanol wears off, you'll spend ten times as long with one tenth of the trouble you might have had. So make it easy for yourself now and later you'll be glad that you aren't worse."

The sky blackened, and Channing knew that they were free in space. Give them another fifteen minutes and the devil himself couldn't find them. With no flight plan scheduled and no course posted, they might as well have been in the seventeenth dimension. As they emerged from the thin atmosphere, there was a fleeting flash of fire from several miles to the East, but Channing did not pay particular attention to it. Arden looked through a telescope, and

said that she thought that she saw a spaceship circling, but that she could not be sure.

Whatever it was, nothing came of it.

The trip out to the Station was a monotonous series of uneventful hours, proceeding along one after the other. They dozed and slept most of the time, eating sparingly and doing nothing that was not absolutely necessary.

Turnabout was accomplished and then the deceleration began, equally long and equally monotonous. It was equally inactive. Channing tried to plan, but it failed because he could not plan without talking and discussing the affair with his men; too much depended upon their co-operation. He fell into a morose, futile feeling that made itself evident in grouching; Arden tried to jolly him, but Don's usually bubbling spirit was doused too deep. Also, Arden herself was none too happy, and she failed to convince herself, which is necessary before one can convince anyone else of anything.

Then they sighted the Station, and Channing's ill spirit left. A man of action, what he hated most was the no-action business of just sitting in a little capsule of steel waiting for the Relay Station to come up out of the sky below. Once it was sighted, Channing could foresee action, and his grouching stopped.

They zipped past the Station at a distance of ten miles, and Channing opened the radio.

"Walt Franks! Wake up, you slumberhead."

The answer came inside of a half minute. "Hello, Don. Who's asleep?"

"Where are you? In Joe's?"

"Joe has declared a drought for the duration," said Franks with a laugh. "He thinks we can't think on Scotch."

"We can't. Have you seen the boys?"

"Murdoch's crew? Sure, they're circling at about five miles, running around in the plane of the ecliptic. Keep running on the colure and the chances are

that you won't even see 'em. But, Don, they can hear us!"

"How about the landing stage at the South end?"

"There are two of them running around the Station at different heights from South to North. The third is circling in a four-mile circle on a plane five miles South of the Station. We've picked up a few HE shells, and I guess that, if you try to make a landing there, you'll be shot to bits. That devil is using the meteor detector for a gun pointer."

"Walt, remember the range finder?"

"Y'mean the one we used to find the *Empress*?"

"Uh-huh. Rig it without the adjustable mirror. Get me? D'you know what I want to do?"

"Yop. All we have to do is to clear away some of the saw grass again. Not too much, though, because it hasn't been too long since we cut it before. I get you all right."

"Fine. How soon?"

"I'm in the beam control dome North. I've got a portable mike, and I walk over to the adjustable mirror and begin to tinker with the moving screws. *Ouch!* I've skun me a knuckle. Now look, Don, I'm going inside and crack the passage end. I've broadcast throughout the Station that this is to be cracked, and the men are swarming all over the axis of the Station doing just that. Come a-running!"

Channing circled the little ship high to the North and came down toward the axis of the Station. He accelerated fiercely for a portion of the time, and then made a slam-bang turnabout that caused some of the plates to complain. A pilot light on the instrument panel gleamed, indicating that some of the plates were strained and that the ship was leaking air. Another light lit, indicating that the automatic pressure control was functioning, and that the pressure was maintained, though it might not be too long.

Then in deceleration, Channing fought the ship on to a die-straight line with the open door at the North end. He fixed the long, long passageway in the

center of his sights, and prayed.

The ship hit the opening squarely, and only then did their terrific speed become apparent. Past bulkhead after bulkhead they drove, and a thin scream came to their ears as the atmosphere down in the bowels of the Station was compressed by the tiny ship's passage.

Doors slammed behind the ship as it passed, and air locks were opened, permitting the Station's center to fill to its normal pressure once more.

Then the rocketing ship slowed. Channing saw a flash of green and knew that the Martial saw grass was halfway down the three-mile length of the Station. He zipped past storeroom and rooms filled with machinery, and then the ship scraped lightly against one of the bulkheads.

It caromed from this bulkhead against the next, hitting it in a quartering slice. From side to side the ship bounced, crushing the bulkheads and tearing great slices from the flanks of the ship.

It slowed, and came to rest against a large room full of packing cases, and was immediately swarmed over by the men from the Relay Station.

They found Channing partly conscious. His nose was bleeding, but otherwise he seemed all right. Arden was completely out, though a quick check by the Station's medical staff assured Don that she would be all right as soon as they gave her a work-out. He was leaving the center of the Station when Franks came puffing up the stairway from the next lowest level.

"Gosh," he said. "It's a real job trying to guess where you stopped. I've been hitting every hundred feet and asking. Well, that was one for the book."

"Yeah," groaned Don. "Come along, Walt. I want a shower. You can give the résumé of the activities whilst I'm showering and trying to soak this adhesive off. Arden, lucky girl, will be unconscious when the doc rips it off, but I never liked the way they remove tape."

"There isn't much to tell," said Franks. "But what there is, I'll tell you."

Channing was finishing the shower when Walt mentioned that it was too bad that they hadn't started his electron gun a few weeks sooner.

Don shut off the water, fumbled for a towel, and said: "What?"

Franks repeated.

Again Channing said: "What? Are you nuts?"

"No. I've been tinkering with an idea of mine. If we had another month to work on it, I think we might be able to clip Murdoch's ears."

"Just what are you using in this super weapon, chum?"

Franks explained.

"Mind if I put in an oar?" asked Channing.

"Not at all. So far we might be able to fry a smelt at twenty feet, or we could cook us a steak. But I haven't been able to do a thing yet. We had it working once, and I think we heated a meteor somewhat, but the whole thing went blooey before we finished the test. I've spent the last week and a half fixing the thing up again, and would have tried it out on the next meteor, but your message brought a halt to everything but cleaning up the mess and making ready just in case we might think of something practical."

"I'll put in my first oar by seeing the gadget. Wait 'til I find my pants, and I'll go right along."

Don inspected the installation and whistled. "Not half bad, sonny. Not half bad."

"Except that we haven't been able to make it work."

"Well, for one thing, you've been running on the wrong track. You need more power."

"Sure," grinned Walt. "More power, he says. I don't see how we can cram any more soup into this can. She'll melt."

"Walt, what happens in a big gun?"

"Powder burns; expanding products of combustion push—"

"Functionally, what are you trying to accomplish? Take it on the basis of a solid shot, like they used to use back in the sailing ship days."

"Well," said Walt thoughtfully, "I'd say they were trying to heave something large enough to do damage."

"Precisely. Qualifying that statement a little, you might say that the projectile transmits the energy of the powder charge to its objective."

"Right," agreed Walt.

"And it is possible to transmit that energy mechanically. I think if we reason this idea out in analogy, we might be able to do it electrically. First, there is the method. There is nothing wrong with your idea, functionally. Electron guns are as old as radio. They—"

The door opened and Arden entered. "Hi, fellows," she said. "What's cooking?"

"Hi, Arden. Like marriage?"

"How long do people have to be married before people stop asking that darned fool question?" asked Arden.

"O.K., how about your question?"

"I meant that. I ran into Warren who told me that the brains were down here tinkering on something that was either a brilliant idea or an equally brilliant flop—he didn't know which. What goes?"

"Walt has turned Buck Rogers and is now about to invent a ray gun."

"No!"

"Yes!"

"Here's where we open a psychopathic ward," said Arden sadly. "So far, Venus Equilateral is the only community that hasn't had a village idiot. But no longer are we unique. Seriously, Walt?"

"Sure enough," said Channing. "He's got an idea here that may work with a little tinkering."

"Brother Edison, we salute you," said Arden. "How does it work?"

"Poorly. Punk. Lousy."

"Well, sound recording has come a long way from the tinfoil cylinder that scratches out: 'Mary had a little lamb!' And transportation has come along swell from the days of sliding sledges. You may have the nucleus of an idea, Walt. But I meant its operation instead of its efficiency."

"We have an electron gun of super size," explained Walt. "The cathode is a big affair six feet in diameter and capable of emitting a veritable storm of electrons. We accelerate them by means of properly spaced anodes of the proper voltage level, and we focus them into a nice bundle by means of electrostatic lenses—"

"Whoa, Tillie. You're talking like the Venerable Buck himself. Say that in language, please."

"Well, we work at swords' points whenever we try to accelerate electrons to high speeds and focus them at the same time," said Walt. "A voltage gradient will cause electrons to change their course like a lens, and with the usual trouble with inanimate things, it works the wrong way. In order to accelerate the electrons emitted from the cathode, they are subjected to the attraction of an anode which is operating at a high positive potential. The electrons leave the cathode, and as they are attracted by the positive anode, they begin to move.

"Now it is common to speak of the velocity of an electron by stating the quantity of volts that the electron has fallen through. The higher the voltage difference between cathode and anode, the faster the electron will go.

"But the stream of electrons will be diverged if it falls through a field of increasing positive potential. In order to bring the stream to a focus, we must follow the first anode with another anode of a potential less than the first anode but still higher than the cathode. That bollixes up the works. It focuses the stream and slows it at the same time. So we follow this anode with another high positive electrode and speed them up again, and then focus them and so on until we get the required velocity. These anodes are shaped like rings so that their electrostatic effect will exist in the center of them; the beam passes through this center."

"In other words the ring-shaped electrodes are electrostatic lenses?"

"Nope. It is the space between them. The lens is either convex or concave depending upon whether the voltage gradient is from positive to less positive or if it is positive to more positive, respectively. In an electrostatic lens for electrons, the thing is not like a glass lens for light. Whether your lens is diverging or converging depends upon which way your electron stream is running through it. With light, a convex lens will converge the light no matter which direction the light is coming from."

"Uh-huh. I see in a sort of vague manner. Now, fellows, go on from there. What's necessary to make this dingbat tick?"

"I want to think out loud," said Channing.

"That's nothing unusual," said Arden. "Can't we get into Joe's? You can't think without a tablecloth, either."

"What I'm thinking is this, Walt. You've been trying to squirt electrons like a fireman runs a hose. Walt, how long do you suppose a sixteen-inch rifle would last if the explosives were constantly replaced and the fire burned constantly?"

"Not long," admitted Walt.

"A gun is an overloaded machine," said Don. "Even a little one. The life of a gun barrel is measured in seconds; totaling up the time of transit of all the rounds from new barrel to worn gun gives a figure expressed in seconds. Your electron gun, Walt, whether it be fish, flesh, or fowl, must be overloaded for an instant."

"Is overload a necessary requirement?" asked Arden. "It seems to me that you might be able to bore a sixteen-inch gun for a twenty-two. What now, little man?"

"By the time we get something big enough to do more than knock paint off, we'll have something bigger than a twenty-two," grinned Channing. "I was speaking in terms of available strength versus required punch. In the way that a girder will hold tremendous overloads for brief instants, a gun is extremely overloaded for milliseconds. We'll have a problem—"

"O.K., aside from that, have you figured out why I haven't been able to do more than warm anything larger than a house brick?"

"Sure," laughed Channing. "What happens in a multigrid radio tube when the suppressor grid is hanging free?"

"Charges up and blocks the electron stream ... hey! That's it!"

"What?" asked Arden.

"Sure," said Walt. "We fire off a batch of electrons, and the first contingent that arrives charges the affair so that the rest of the beam sort of wriggles out of line."

"Your meteor is going to take on a charge of phenomenal negative value, and the rest of your beam is going to be deflected away, just as your electron lenses deflect the original beam," said Channing. "And now another thing, old turnip. You're squirting out a lot of electrons. That's much amperage. Your voltage—velocity—is nothing to rave about. Watts is what you want, to corn a phrase."

"*Phew*," said Walt. "Corn, he says. Go on, prodigy, and make with the explanations. I agree, we should have more voltage and less quantity. But we're running the stuff at plenty of voltage now. Nothing short of a Van Der Graf generator would work—and while we've got one up on the forty-ninth level, we couldn't run a supply line down here without reaming a fifty-foot hole through the Station, and then I don't know how we'd get that kind of voltage down here without ... that kind of stuff staggers the imagination. You can't juggle a hundred million volts on a wire. She'd squirt off in all directions."

"Another thing, whilst I hold it in my mind," said Channing thoughtfully. "You go flinging electrons off the Station in basketful after basketful, and the next bird that drops a ship on the landing stage is going to spot-weld himself right to the South end of Venus Equilateral. It wouldn't be long before the Station would find itself being pulled into Sol because of the electrostatic stress—if we didn't run out of electrons first!"

"I hardly think that we'd run out—but we might have a tough time flinging them away after a bit. Could be that we should blow out a fist full of positrons at the same time?"

"Might make up a concentric beam and wave the positive ions at the target," said Channing. "Might help."

"But this space-charge effect. How do we get around that?"

"Same way we make the electron gun work. Fire it off at a devilish voltage. Run your electron velocity up near the speed of light; the electrons at that speed will acquire considerable mass, in accordance with Lorenz's equation which shows that as the velocity of a mass reaches the speed of light, its

mass becomes infinite. With a healthy mass built up by near-light velocities, the electrons will not be as easy to deflect. Then, too, we can do the damage we want before the charge can be built up that will deflect the stream. We ram 'em with a bundle of electrons moving so fast that the charging effect can not work; before the space charge can build up to the level required for self-nullification of our beam, the damage is done."

"And all we need is a couple of trillion volts. Two times ten to the twelfth power. *Grrr*."

"I can see that you'll need a tablecloth," said Arden. "You birds can think better over at Joe's. Come along and feed the missus, Don."

Channing surveyed the instrument again, and then said: "Might as well, Walt. The inner man must be fed, and we can wrangle at the same time. Argument assists the digestion—and vice versa."

"Now," said Channing, as the dishes were pushed aside, clearing a space on the table. "What are we going to do?"

"That's what I've been worrying about," said Walt. "Let's list the things that make our gun ineffective."

"That's easy. It can't dish out enough. It's too dependent upon mobility. It's fundamentally inefficient because it runs out of ammunition too quick, by which I mean that it is a sort of gun with antiseptic bullets. It cures its own damage."

"Prevents," said Arden.

"All right, it acts as its own shield, electrostatically."

"About this mobility," said Walt, "I do not quite agree with that."

"You can't whirl a hunk of tube the size and weight of a good-sized telescope around fast enough to shoot holes in a racing spaceship," said Channing. "Especially one which is trying to dodge. We've got to rely upon something that can do the trick better. Your tube did all right following a meteor that runs in a course that can be predicted, because you can set up your meteor spotter to correct for the mechanical lag. But in a spaceship that is trying to

duck your shot, you'll need something that works with the speed of light. And, since we're going to be forced into something heavy and hard-hitting, its inertia will be even more so."

"Heavy and hard-hitting means exactly what?"

"Cyclotron or betatron. One of those dingusses that whirls electrons around like a stone on a string until the string breaks and sends the stone out at a terrific speed. We need a velocity that sounds like a congressional figure."

"We've got a cyclotron."

"Yeah," drawled Channing. "A wheezy old heap that cries out in anguish every time the magnets are charged. I doubt that we could move the thing without it falling apart. The betatron is the ticket."

"But the cyclotron gives out with a lot more soup."

"If I had to increase the output of either one, I could do it a lot quicker with the betatron," said Channing. "In a cyclotron, the revolution of the electrons in their acceleration period is controlled by an oscillator, the voltage output of which is impressed on the D chambers. In order to speed up the electron stream, you'd have to do two things. One: Build a new oscillator that will dish out more power. Two: Increase the strength of the magnets."

"But in the betatron, the thing is run differently. The magnet is built for A. C. and the electron gun runs off the same. As your current starts up from zero, the electron gun squirts a bouquet of electrons into a chamber built like a pair of pie plates set rim to rim. The magnet's field begins to build up at the same time, and the resulting increase in field strength accelerates the electrons and at the same time, its increasing field keeps the little devils running in the same orbit. Shoot it with two-hundred-cycle current, and in the half cycle your electrons are made to run around the center a few million times. That builds up a terrific velocity—measured in six figures, believe it or not. Then the current begins to level off at the top of the sine wave, and the magnet loses its increasing phase. The electrons, still in acceleration, begin to whirl outward. The current levels off for sure and begins to slide down—and the electrons roll off at a tangent to their course. This stream can be collected and used. In fact, we have a two-hundred-cycle beam of electrons at a couple of billion volts. That, brother, ain't hay!"

"Is that enough?"

"Nope."

"Then how do you hope to increase this velocity? If it is easier to run this up than it would be the cyclotron, how do we go about it?"

Channing smiled and began to draw diagrams on the tablecloth. Joe looked over with a worried frown, and then shrugged his shoulders. Diagrams or not, this was an emergency—and besides, he thought, I need another lesson in high-powered gadgetry.

"The nice thing about this betatron," said Channing, "is the fact that it can and does run both ends on the same supply. The current and voltage phases are correct so that we do not require two supplies which operate in a carefully balanced condition. The cyclotron is one of the other kinds; though the one supply is strictly D.C., the strength of the field must be controlled separately from the supply to the oscillator that runs the D plates. You're sitting on a fence, juggling knobs and stuff all the time you are bombarding with a cyc.

"Now let us inspect the supply of the betatron. It is sinusoidal. There is the catch. There is the thing that makes it possible. That single fact makes it easy to step the power up to terrific quantities. Since the thing is fixed by nature so that the input is proportional—electron gun initial velocity versus magnetic field strength, if we increase the input voltage, the output voltage goes up without having to resort to manipulative gymnastics on the part of the operator."

"Go on, Professor Maxwell."

"Don't make fun of a great man's name," said Arden. "If it wasn't for Clerk Maxwell, we'd still be yelling out of the window at one another instead of squirting radio beams all over the Solar System."

"Then make him quit calling me Tom Swift."

"Go on, Don, Walt and I will finish this argument after we finish Hellion Murdoch."

"May I?" asked Channing with a smile. He did not mind the interruption; he was used to it in the first place and he had been busy with his pencil in the second place. "Now look, Walt, what happens when you smack a charged condenser across an inductance?"

"You generate a damped cycle of the amplitude of the charge on the condenser, and of frequency equal to the L, C , constants of the condenser and inductance. The amplitude decays according to the factor Q , following the equation for decrement—"

"Never mind, I've got it here on my whiteboard," smiled Channing, pointing to the tablecloth. "You are right. And the purity of the wave?"

"Sinusoidal ... hey! That's it!" Walt jumped to his feet and went to the telephone.

"What's 'it'?" asked Arden.

"The betatron we have runs off of a five-hundred-volt supply," chuckled Channing. "We can crank that up ten to one without running into any difficulty at all. Five-hundred-volt insulation is peanuts, and the stuff they put on wires nowadays is always good for ten times that just because it wouldn't be economical to try to thin the insulation down so that it only protects five hundred. I'll bet a hat that he would crank the input up to fifty thousand volts without too much sputtering—though I wouldn't know where to lay my lunch hooks on a fifty-thousand-volt condenser of any appreciable capacity. Well, stepping up the rig ten to one will dish us out just shy of a couple of thousand million volts, which, as brother Franks says, is not hay!"

Walt returned after a minute and said: "Warren's measuring the inductance of the betatron magnet. He'll calculate the value of C required to tune the thing to the right frequency and start to achieve that capacity by mazing up whatever high-voltage condensers we have on the Station. Now, Don, let's calculate how we're going to make the thing mobile."

"That's a horse of a different color. We'll have to use electromagnetic deflection. From the constants of the electron stream out of our souped-up Suzy, we'll have to compute the necessary field to deflect such a beam.

That'll be terrific, because the electrons are hitting it up at a velocity approaching that of light—maybe a hundred and seventy thousand miles per—and their mass will be something fierce. That again will help to murder Murdoch; increasing mass will help to keep the electrons from being deflected, since it takes more to turn a heavy mass—et cetera, see Newton's laws of inertia for complete statement. Have 'em jerk the D plates out of the cyc and bring the magnet frame down here—to the turret, I mean—and set 'em up on the vertical. We'll use that to run the beam up and down, we can't possibly get one hundred and eighty degree deflection, of course, but we can run the deflection over considerable range. It should be enough to catch a spaceship that is circling the Station. For the horizontal deflection, what have we got?"

"Nothing. But the cyc magnet is a double pole affair. We could break the frame at the D plates and set one winding sidewise to the other and use half on each direction."

"Sure. Have one of Warren's gang fit the busted pole pieces up with a return-magnetic frame so that the field will be complete. He can weld some girders on and around in an hour. That gives us complete deflection properties left and right; up and down. We should be able to cover a ninety-degree cone from your turret."

"That'll cover all of Murdoch's ships," said Walt.

"Too bad we haven't got some U_{235} to use. I'd like to plate up one of his ships with some positive ions of U_{235} and then change the beam to slow neutrons. That might deter him from his life of crime."

"Variations, he wants," said Arden. "You're going to impale one ship on a beam of electrons, one ship on a beam of U_{235} ions; and what will you have on the third?"

"I'll think of something," said Channing. "A couple of pounds of U_{235} should make things hum, though."

"More like making them disappear," said Franks. "*Swoosh!* No ship. Just an incandescent mass falling into the Sun. I'm glad we haven't got U_{235} in any quantity out here. We catch a few slow neutrons now and then, and I wouldn't be able to sleep nights. The things just sort of wander right through

the Station as though it weren't here at all; they stop just long enough to register on the counter upstairs and then they're gone."

"Well, to work, people. We've got a job to do in the next three and a half days."

Those days were filled with activity. Hauling the heavy parts down to the turret was no small job, but it was accomplished after a lot of hard work and quite a bit of tinkering with a cutting torch. The parts were installed in the outer skin, and the crew with the torch went back over their trail and replaced the gaping holes they left in the walls and floors of Venus Equilateral. The engineering department went to work, and for some hours the place was silent save for the clash of pencil on paper and the scratching of scalp. The most popular book in the Station became a volume on nuclear physics, and the second most popular book was a table of integrals. The stenographic force went to work combing the library for information pertaining to electronic velocities, and a junior engineer was placed in as buffer between the eager stenographers and the harried engineering department. This was necessary because the stenographers got to the point where they'd send anything at all that said either "electrons" or "velocity," and one of the engineers read halfway through a text on atomic structure before he realized that he had been sold a bill of goods. Wire went by the mile down to the turret, and men proceeded to blow out half of the meters in the Station with the high-powered beam. Luckily, the thing was completely nonspectacular, or Murdoch might have gained an inkling of their activities. The working crew manipulated constants and made haywire circuits, and finally announced that the beam would deflect—if the calculations were correct.

"They'd better be," said Channing. He was weary. His eyes were puffed from lack of sleep, and he hadn't had his clothing off in three days.

"They are," said Franks. He was in no better shape than Don.

"They'd better be right," said Channing ominously. "We're asking for a kick in the teeth. The first bundle of stuff that leaves our gun will energize Murdoch's meteor spotter by shear electrostatic force. His gun mounts,

which you tell me are coupled to the meteor detector for aiming, will swivel to cover the turret out here. Then he'll let us have it right in the betatron. If we don't get him first, he'll get us second."

"Don," said Walt in a worried voice, "How are we going to replace the charge on the Station? Like the bird who was tossing baseballs out of the train—he quit when he ran out of them. Our gun will quit cold when we run out of electrons—or when the positive charge gets so high that the betatron can't overcome the electrostatic attraction."

"Venus Equilateral is a free grid," smiled Channing. "As soon as we shoot off electrons, Old Sol becomes a hot cathode and our Station collects 'em until the charge is equalized again."

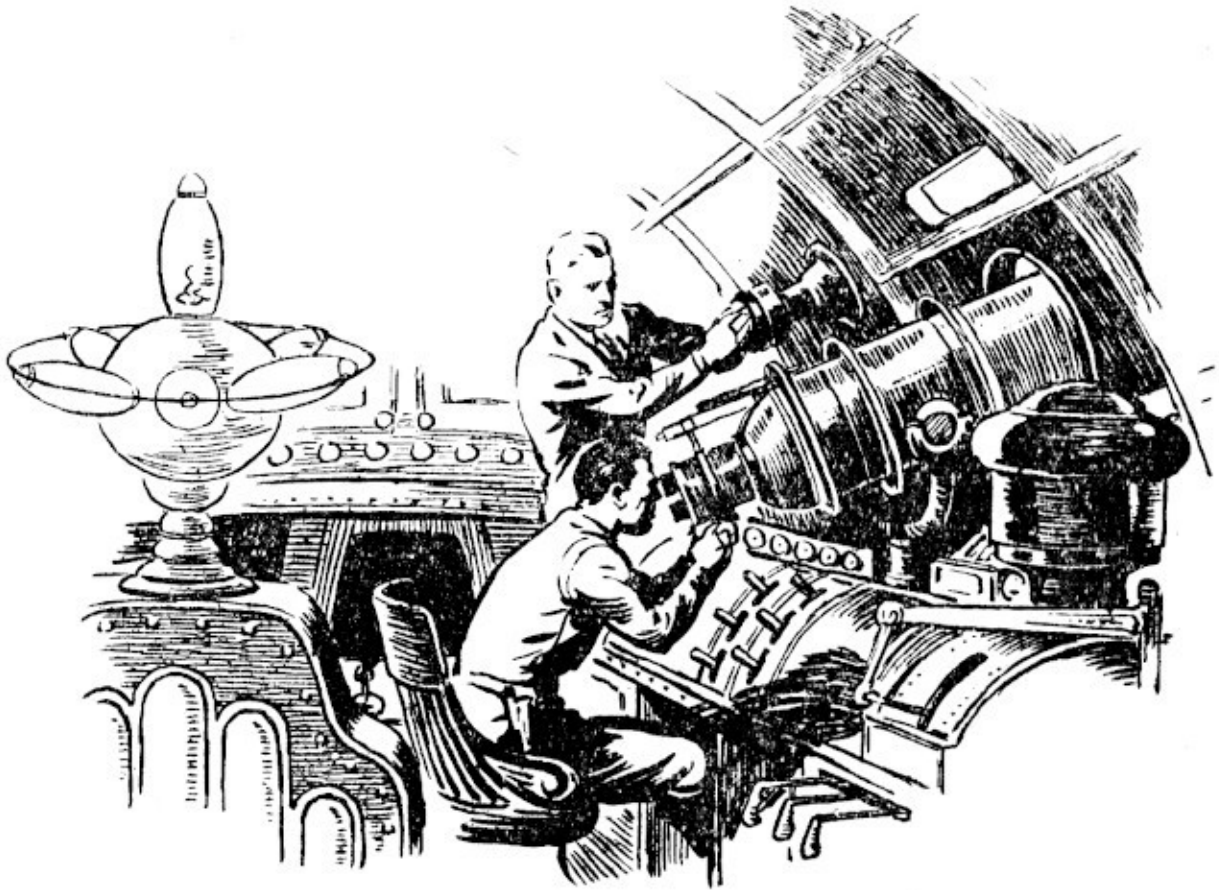
"And what happens to the bird who is holding on to something when we make off with a billion volts? Does he scrape himself off the opposite wall in a week or so—after he comes to—or can we use him for freezing ice cubes? Seems to me that it might be a little bit fatal."

"Didn't think of that," said Channing. "There's one thing, their personal charge doesn't add up to a large quantity of electricity. If we insulate 'em and put 'em in their spacesuits, they'll be all right as long as they don't try to grab anything. They'll be on the up and down for a bit, but the resistance of the spacesuit is high enough to keep 'em from draining all their electrons out at once. I recall the experiments with early Van Der Graf generators at a few million volts—the operator used to sit in the charged sphere because it was one place where he couldn't be hit by man-made lightning. It'll be rough, but it won't kill us. Spacesuits, and have 'em sit in plastic chairs the feet of which are insulated from the floor by china dinner plates. This plastic wall covering that we have in the apartments is a blessing. If it were all bare steel, every room would be a miniature hell. Issue general instructions to that effect. We've been having emergency drills for a long time, now's the time to use the grand collection of elastomer spacesuits. Tell 'em we give 'em an hour to get ready."

Hellion Murdoch's voice came over the radio at exactly the second of the expiration of his limit. He called Channing and said:

"What is your answer, Dr. Channing?"

Don squinted down the pilot tube of the meteor spotter and saw the *Hippocrates* passing. It was gone before he spoke, but the second ship came along, and the pilot tube leaped into line with it. Don checked meters on the crude panel before him, and then pressed the plastic handle of a long lever.



There was the crash of a heavy-duty oil switch.

That was all.

Crackles of electricity flashed back and forth through the Station, and the smell of ozone arose. Electric light filaments leaned over crazily, trying to touch the inner walls of the glass. Panes of glass ran blue for an instant, and the nap of the carpets throughout the Station stood bolt upright. Hair stood on end, touched the plastic helmet dome, discharged, fell to the scalp, raised again and discharged, fell once more, and then repeated this raising and falling, again and again and again. Electric clocks ran crazily, and every bit of electronic equipment on the Station began to act in an unpredictable manner.

Then things settled down again as the solar emission charged the Station to equilibrium.

Aboard the ship, it was another story. The celestial globe of the meteor spotter blazed once in a blinding light and then went completely out of control. It danced with pin points of light, and the coupler that was used to direct the guns went crazy. Turrets tried to swivel, but the charge raised hob with the electronic controls, and the guns raised once and then fell, inert. One of them belched flame and fire, and the shell went wild. The carefully balanced potentials in the driver tubes was upset, and the ship lost headway. The heavy ion stream from the driving cathode bent and spread, touching the dynodes in the tube. The resulting current brought them to a red heat, and they melted down and floated through the evacuated tube in round droplets. Instruments went wild, and gave every possible answer, and the ship became a bedlam of ringing bells and flashing danger lights.

But the crew was in no shape to appreciate the display. From metal parts in the ship there appeared coronas that reached for the unprotected men, and seared their flesh. And since their gravity-apparent was gone, they floated freely through the air, and came in contact with highly charged walls, ceiling, and floor; to say nothing of the standard metal furniture.

It was a sorry bunch of pirates that found themselves in a ship-without-motive-power that was beginning to leave their circular course on a tangent that would let them drop into the Sun.

"That's my answer, Murdoch!" snapped Channing. "Watch your second ship!"

"You young devil," shouted Murdoch, "what did you do?"

"You never thought that it would be an electronics engineer that made the first energy gun, did you, Murdoch? I'm now going to take a shot at No. 3!"

No. 3's turrets swiveled around and from the guns flashes of fire came streaming. Channing punched his lever savegely, and once again the Station was tortured by the effects of its own offensive.

Ship No. 3 suffered the same fate as No. 2.

Then, seconds later, armor-piercing shells began to hit Venus Equilateral. They hit, and because of the terrific charge, they began to arc at the noses. The terrible current passed through the fuses, and the shells exploded on contact instead of boring inside before detonation. Metal was bent and burned, but only a few tiny holes resulted. As the charge on the Station approached equilibrium once more, men ran with torches to seal these holes.

"Murdoch," said Channing, "I want you!"

"Come and get me."

"Land—or die!" snapped Channing in a vicious tone. "I'm no humanitarian, Murdoch. You'd be better off dead!"

"Never," said Hellion Murdoch.

Channing punched the lever for the third time, but as he did, Murdoch's ship leaped forward under several G. The magnets could not change in field soon enough to compensate for this change in direction, and the charge failed to connect as a bull's-eye. It did expend some of its energy on the tail of the ship. Not enough to cripple the vessel, but the *Hippocrates* took on a charge of enough value to make things hard on the crew.



Metal sparked, and instruments went mad. Meters wound their needles against the end pegs. The celestial globe glinted in a riot of color and then went completely dead. Gun servers dropped their projectiles as they became too heavily charged to handle, and they rolled across the turret floors, creating panic in the gun crews. The pilot fought the controls, but the charge on his driver tubes was sufficient to make his helm completely unpredictable. The panel sparked at him and seared his hands, spoiling his nervous control and making him heavy-handed.

"Murdoch," cried Channing in a hearty voice, "that was a miss! Want a hit?"

Murdoch's radio was completely dead. His ship was yawing from side to side as the static charges raced through the driver tubes. The pilot gained

control after a fashion, and decided that he had taken enough. He circled the Station warily and began to make a shaky landing at the South end.

Channing saw him coming, and with a glint in his eye, he pressed the lever for the fourth and last time.

Murdoch's ship touched the landing stage just after the charge had been driven out into space. The heavy negative charge on the *Hippocrates* met the heavy positive charge on Venus Equilateral. The ship touched, and from that contact, there arose a cloud of incandescent gas. The entire charge left the ship at once, and through that single contact. When the cloud dissipated, the contact was a crude but efficient welded joint that was gleaming white-hot.

Channing said to Walt: "That's going to be messy."

Inside of the *Hippocrates*, men were still frozen to their handholds. It was messy, and cleaning up the *Hippocrates* was a job not relished by those who did it.

But cleaning up Venus Equilateral was no small matter either.

Weeks went by before the snarled-up instruments were repaired. Weeks in which the captured *Hippocrates* was repaired, too, and used to transport material and special supplies from Terra, and Venus, and Mars. Weeks in which the service from planet to planet was interrupted and erratic.

Then one day, service was restored, and life settled down to a reasonable level. It was after this time that Walt and Channing found time to spend an idle hour together. Walt raised his glass and said: "Here's to electrons!"

"Yeah," grinned Channing. "Here's to electrons. Y'know, Walt, I was a little afraid that space might become a sort of wild West show, with the ships bristling with space guns and betatrons and stuff like that. In which case you'd have been a stinking benefactor. But if the recoil is as bad as the output—and Newton said that it must be—I can't see ships cluttering up their insides with stuff that'll screw up their instruments and driver tubes. But the thing that amuses me about the whole thing is the total failure you produced."

"Failure?" asked Walt. "What failed?"

"Don't you know? Have you forgotten? Do you realize that spaceships are still ducking around meteors instead of blasting them out of the way with the Franks Electron Gun? Or did you lose sight of the fact that this dingbat started out in life as a meteor-sweeper?"

Walt glared over the rim of his glass, but he had nothing to say.

THE END.

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